SAMARJEET MALIK

+91 9315932844 | samarjeetmalik.gs5@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

Indian Institute of Technology (IIT) Jodhpur — Dec 2024 – Feb 2026
Predoctoral Fellowship, Artificial Intelligence and Data Science — **GPA: 3.9/4.0**

Kalinga Institute of Industrial Technology (KIIT) — Oct 2021 – Sep 2025
B.Tech, Computer Science and Engineering — **CGPA: 9.15/10**

The Shriram Millennium School, Faridabad — 2021
ISC (Class XII – Science) — **94.25%**

COURSEWORK

Computer Science & Engineering Core: Data Structures and Algorithms, Object Oriented Programming, Operating Systems, Database Management Systems, Computer Networks, Computer Organization and Architecture, Automata and Formal Languages, Compiler Design, Software Engineering, Web Technology, Design and Analysis of Algorithms, Software Project Management, Cloud Computing, High Performance Computing

Mathematics, Physics & Scientific Computing: Mathematics I & II, Differential Geometry, Computational Physics, Discrete Mathematics, Probability & Statistics, Numerical Methods, Physics, Chemistry, Biology, Environmental Science, Engineering Graphics, Basic Manufacturing Systems

Artificial Intelligence, Machine Learning & Data Science: Machine Learning, Artificial Intelligence, Computational Intelligence, Natural Language Processing, What is Data Science?, Data Analysis with Python

Electronics, Systems & Professional Studies: Basic Electrical Engineering, Analog Electronic Circuits, Digital Electronics, Electrical Circuits Analysis, Principle of Digital Communication, Engineering Economics, Human Resource Management, Professional Communication, Business Communication, Professional Practice Law & Ethics, Yoga and Human Consciousness, Astrophysics

RESEARCH EXPERIENCE

Indian Institute of Technology (IIT) Jodhpur — Dec 2024 – Feb 2026
Predoctoral Fellow – Physics-Informed Machine Learning — Jodhpur, India

- Developed **Physics-Informed Neural Networks (PINNs)** for cosmic-ray energy reconstruction, training on **171,776 simulated air-shower events** using GPU-accelerated pipelines; embedded air-shower physics constraints (electromagnetic scaling, muon multiplicity, zenith monotonicity) as soft regularization on network manifolds.
- Derived **curvature analysis of muon scaling surfaces** in logarithmic energy-mass space, extracting hadronic multiplicity growth exponent $\delta = 0.077 \pm 0.014$ via Gaussian curvature eigenvalues and hyperbolic geometry.
- Built **ConFIG (Conflict-Free Inverse Gradient)** framework for multi-objective PINN training, resolving gradient conflicts via geometric projection on loss manifolds; achieved 30% faster convergence and 25% improved stability in high-fidelity physics simulations.
- **PIXEL/SPINN Project:** Developing grid-based Physics-Informed Local feature Networks (PIXEL) for shape optimization, targeting Paul Trap electrode multipole optimization ($|a_4/a_2| < 10^{-4}$) and C-shape magnetostatics; implementing multi-resolution training (32→64→128) with feature interpolation.
- **Automated Asteroid Detection Pipeline:** Architected 9-module, 3,400-line Python framework for International Asteroid Search Campaign (IASC) with MPC-formatted output; implemented dipole detection, tracklet building with 4-D DBSCAN, Gauss orbit fitting, and injection-recovery completeness testing (50% limiting magnitude estimation).
- **PAI26 Submission:** Preparing manuscript on PIXEL architecture for Stanford Physics and AI Conference (PAI26), demonstrating 529× interface loss improvement over MLP baselines and validation across Helmholtz, Burgers', and Laplace PDEs.
- Benchmarked against XGBoost/DNN baselines; manuscript in preparation for **NeurIPS/ICML**.

Tata Institute of Fundamental Research (TIFR) – GRAPES-3 — Apr 2025 – Feb 2026
Machine Learning Research Intern – Astroparticle Physics — Ooty, India

- Built **equivariant CNN surrogate models** for extensive air-shower classification, achieving **96.8% accuracy** with **75 ms GPU inference latency**; incorporated **rotational symmetry constraints** for robustness in detector arrays.
- Designed physics-aware preprocessing and robustness evaluations, delivering 24% SNR improvement under atmospheric and energy-domain distribution shifts.

Bhabha Atomic Research Centre (BARC) — Jul 2024 – Nov 2024
Applied Physics Research Intern — Mumbai, India

- Developed **GPU-accelerated PINN surrogates** for COMSOL multiphysics simulations, solving **200+ high-dimensional PDE systems** with symmetry-preserving constraints for improved generalization.
- Validated models using **Beer–Lambert spectroscopy** and Monte Carlo uncertainty propagation, achieving **98.7% accuracy** across **10,000+ samples** with calibrated epistemic uncertainty.

Defence Research and Development Organization (DRDO) — Dec 2023 – Apr 2024

AI–ML Research Intern Pune, India

- Integrated **GPR and thermal sensing** with deep learning for **DRDO's Daksh robot**, enabling detection of plastic-lined and low-metallic landmines.
- Designed **CNN and Siamese architectures** with GAN-based enhancement, achieving **98.74% accuracy**, **95.11% F1**, and **0.92 mAP**.

National Institute for Transforming India (NITI Aayog) Dec 2024 – Present
Research Intern – Policy & Economics New Delhi, India

- Conducted policy research for 3+ national projects across Economics, Finance, Trade, and G20 verticals, supporting IAS/IRS officers in evidence-based decision-making.
- Contributed to research-backed policy reports on economic development, multilateral cooperation, and BRICS summit preparations.

TEACHING, MENTORSHIP & LEADERSHIP

Guest Lecturer, Delhi Technological University (DTU) 2024
Delivered lecture on *Machine Learning Applications in Physics* to undergraduate engineering students.

Chief Advisor, TEDx Sri Ram College of Commerce (SRCC), DU 2023 – 2024
Led curation and mentorship team for university-level TEDx conference; guided speaker selection and content development.

WWF India – Young Conservationist Programme 2020 – 2021
Led conservation outreach and fundraising initiatives; coordinated campaigns on wildlife protection and environmental sustainability.

Student Mentorship Program 2022 – Present
Mentored 4 undergraduate researchers with verified placement outcomes:

- Avishikta Bhattacharjee – ETH Zurich (Masters thesis, Physics)
- Vasu Aggarwal – IIT BHU (Research fellowship)
- Swastik Panda – IIT Kharagpur (Research program)
- Shristy Yadav – CSIR (Research internship)

Team Leadership – Technical Competitions
Led teams in national-level hackathons and research projects:

- **Smart India Hackathon 2023 – Finalist:** Led 5-person team developing AI solution for ISRO challenge
- **ISRO Antriksh Hackathon 2024 – Semifinalist:** Led 5-person team in space technology innovation
- **DRDO:** Led 2 teams of 3 researchers each in robotics and sensing projects
- **BARC:** Led 4-person team in GPU-accelerated simulation development

School Leadership Positions 2019 – 2021

- **School President**: Elected student body head; represented student interests before school administration and led institution-wide initiatives.
- **Head of Sports**: Oversaw inter-house and inter-school sports programs; coordinated teams, schedules, and discipline across multiple cohorts.
- **House Captain (Winning House)**: Led a large student cohort to competitive success; responsible for strategy, coordination, and morale.

SELECTED PROJECTS

Geometric Origin of Curvature in Muon Scaling *Python, PyTorch, Differential Geometry*
• **Independent derivation** of hyperbolic geometry in cosmic-ray air showers: mapped muon multiplicity scaling as **2D Riemannian surface** in $(\log E, \log A, \log N_\mu)$ space.
• Extracted **multiplicity growth exponent** δ via **principal curvature analysis** (Gaussian curvature K, eigenvalues κ_1, κ_2), connecting microscopic QCD to macroscopic shower geometry.

Conflict-Free PINN Training Methods *Python, TensorFlow* GitHub
• Built **Conflict-Free Inverse Gradient (ConFIG)** framework for PINNs, resolving gradient conflicts in multi-objective training via geometric projection on loss manifolds.
• Achieved **30% faster convergence**, **25% improved stability**, and **15% accuracy gains** in high-fidelity physics simulations.

Solving Inverse Physics Problems with Score Matching *Python, PyTorch* GitHub
• Designed score-matching framework for inverse physics problems, learning probability distributions on high-dimensional manifolds via denoising score matching.
• Achieved **95% accuracy**, **0.92 F1**, and **20% faster inference** (5s→4s) for CERN-aligned research.

SPINN: Physics-Informed Neural Network Shape Optimization *Python, PyTorch, FEniCS*
• Extended coordinate-projection framework with **PIXEL** (grid-based representations) and **PIG** (Physics-Informed Gaussians) to eliminate spectral bias in shape optimization.
• Applied to **Paul Trap electrode optimization** achieving 4-rod geometry with anharmonicity $\alpha < 0.01$ and trap depth > 0.5eV for $^{111}Cd^+$ ions; 529× improvement in interface loss over standard MLPs.

PUBLICATIONS

Manuscripts available upon request.

[C.1] Malik, S. (2026). **Geometric Origin of Curvature in Muon Scaling and Extraction of Hadronic Multiplicity Growth.** *Stanford Physics and AI Conference (PAI26), Center for Decoding the Universe.* **Target venue.** – *Derives closed-form mapping between curvature coefficients and QCD multiplicity growth using differential geometry.*

[J.1] Malik, S. (2025). PINN simulation of laser-induced graphene sensors for cancer detection. *Computational Physics (Elsevier).*

[J.2] Malik, S., et al. (2025). AI-enhanced Wi-Fi sensing for human motion detection. *IEEE IoT.*

[C.2] Nayak, M.G., & Malik, S. (2025). **Future Trends in Cybersecurity and Blockchain.** *Springer ICMLCS.* **DOI:** 10.1007/978-3-032-14156-9_37

[C.3] Malik, S., Ghani, A., Nayak, M.G. (2025). **Real Estate Price Predictor: ML Framework for Property Valuation.** *IEEE ICNGTSE.*

[C.4] Malik, S. (2025). **Improving Situational Awareness of Autonomous Vehicles.** *Springer ICMLCS.* **DOI:** 10.1007/978-3-032-14156-9_35

[C.5] Bhattacharjee, A., et al. (2025). Efficiency–Accuracy Tradeoffs in CNNs vs. ViTs. *IEEE ICNGTSE.*

[C.6] Maiti, A., Muley, A., Malik, S. (2025). Murmuration: Multi-Agent Orchestration in Flutter. *IEEE ICNGTSE.*

TECHNICAL SKILLS

Programming:	Python, C++, C, Java, SQL, R, Bash
ML/DL:	PyTorch, TensorFlow, Keras, Scikit-learn, JAX
Mathematics:	Differential Geometry, Algebraic Topology, Group Theory, PDEs, Spectral Methods
Astronomy:	AstroPy, HEALPix, FITS I/O, Astropy Tables, SEP, astroalign, astroquery
Simulation:	COMSOL, Ansys Fluent, FEniCS, Monte Carlo methods, PDE solvers
Visualization:	Matplotlib, Seaborn, Plotly, WebGL, Three.js
HPC/Tools:	Linux, Git, SLURM, CUDA, PostgreSQL, Docker, LaTeX

AWARDS & ACHIEVEMENTS

- **International Mathematics Olympiad (IMO) – 2× Gold Medalist**, International Rank **97**, International Medal of Merit.
- **National Science Olympiad (NSO)** – 2× Gold Medalist, International Rank **248**, Zonal Medal of Merit.
- **SOF National Cyber Olympiad (NCO)** – Gold Medalist, International Rank **148**, Zonal Medal of Excellence.
- **Smart India Hackathon 2023 – Finalist** and **ISRO Antriksh Hackathon 2024 – Semifinalist**.
- **IASC NASA-PanSTARRS Certificate** – Recognized for asteroid detection and astrometric validation.
- **HPAIR Harvard Virtual Conference 2025** – Selected Delegate (competitive review).
- **8 Parliamentary Debate Victories** – BPD/APD formats at NLU Delhi, DU, NLU Patna, NIT Rourkela (Winner & Order of Merit).
- **9 of 11 MUN Wins** – DU, IIT, SSU, RU, JU, KIIT.
- **North Zone Captain** – National Gold Medalist in 110m hurdles and 4×100m relay; Judo Silver Medalist (Haryana, U-17/19).

RECOMMENDATIONS

1. **Dr. Martin Mascarenhas**
   Outstanding Scientist
   Director **BTDG**
   BARC, Mumbai
   Brief: One of the **world's leading researchers** in **Radio Frequency** and **X-Ray** sciences and **Head Scientist** at India's premier institution in **nuclear science** and **atomic energy**.
   Relationship: Research In-Charge

2. **Sanjeet Singh, IRS**
   Senior Adviser
   Eco and Finance – II Division
   NITI Aayog, New Delhi
   Brief: Senior civil servant and **policy adviser** specialising in **macroeconomic policy**, **public finance**, and **institutional governance**. Leads strategic advisory on national economic reforms and fiscal frameworks at India's apex public policy think tank.
   Relationship: Policy Mentor & Supervisor

3. **Bani Hazra**
   Scientific Officer (G)
   Group Head ATRC
   R&DE(Engrs), **DRDO**, Pune
   Brief: Pioneer of the **Daksh DRDO robot** and leading scientist in **Robotics, Sensors, AI,** and **Electronics**. Heads the **AI-Robotics Division** at India's premier defence R&D organisation.
   Relationship: Research In-Charge